G L Bajaj Group of Institutions, Mathura

Worksheet Problems on Data Structures and Algorithms

Worksheet 7

Q1:Reverse Words

Given a String str, reverse the string without reversing its individual words. Words are separated by dots.

Note: The last character has not been '.'.

Examples :

Input: str = i.like.this.program.very.much

Output: much.very.program.this.like.i

Explanation: After reversing the whole string(not individual words), the input string becomes much.very.program.this.like.i

Input: str = pqr.mno

Output: mno.pqr

Explanation: After reversing the whole string , the input string becomes mno.pqr

Q2:Permutations of a given string

Given a string s. The task is to return a vector of string of all unique permutations of the given string, s that may contain dulplicates in lexicographically sorted order.

Examples:

Input: ABC

Output: [ABC, ACB, BAC, BCA, CAB, CBA]

Explanation: Given string ABC has permutations in 6 forms as ABC, ACB, BAC, BCA, CAB and CBA .

Input: ABSG

Output: [ABGS, ABSG, AGBS, AGSB, ASBG, ASGB, BAGS, BASG, BGAS, BGSA, BSAG, BSGA, GABS, GASB, GBAS, GBSA, GSAB, GSBA, SABG, SAGB, SBAG, SBGA, SGAB, SGBA]

Explanation: Given string ABSG has 24 permutations.

Q3: Longest Palindrome in a String

Given a string str, find the longest palindromic substring in str. Substring of string str: str[i . . . . j] where $0 \le i \le j < len(str)$. Return the longest palindromic substring of str.

Palindrome string: A string that reads the same backward. More formally, str is a palindrome if reverse(str) = str. In case of conflict, return the substring which occurs first (with the least starting index).

Examples :

Input: str = "aaaabbaa"

Output: aabbaa

Explanation: The longest Palindromic substring is "aabbaa".

Input: str = "abc"

Output: a

Explanation: "a", "b" and "c" are the longest palindromes with same length. The result is the one with the least starting index.

Q4:Recursively remove all adjacent duplicates

Given a string s, remove all its adjacent duplicate characters recursively.

Note: For some test cases, the resultant string would be an empty string. In that case, the function should return the empty string only.

Example 1:

Input:

S = "geeksforgeek"

Output: "gksforgk"

Explanation:

g(ee)ksforg(ee)k -> gksforgk

Example 2:

Input:

S = "abccbccba"

Output: ""

Explanation:

ab(cc)b(cc)ba->abbba->a(bbb)a->aa->(aa)->""(empty string)

Q5:String Rotated by 2 Places

Given two strings a and b. The task is to find if the string 'b' can be obtained by rotating (in any direction) string 'a' by exactly 2 places.

Example 1:

Input:

a = amazon

b = azonam

Output:

1

Explanation:

amazon can be rotated anti-clockwise by two places, which will make it as azonam.

Example 2:

Input:

a = geeksforgeeks

b = geeksgeeksfor

Output:

0

Explanation:

If we rotate geeksforgeeks by two place in any direction, we won't get geeksgeeksfor.